Junjie (Takku) Li

@ alvinbluy@gmail.com | Portfolio | LinkedIn | GitHub | Seattle (Open to Relocate)

Education

Northeastern University Sep 2023 - Dec 2025
Master of Science in Computer Science Seattle, WA
Teaching Assistant: Object Oriented Design, Algorithms

Nankai University Tianjin, China
B.S. in Mathematics

Experience

Berkshire Hathaway HomeServices: Machine Learning Engineer Intern Jun 2024 - Aug 2024

- Customized and fine-tuned offline **deep learning models** (N-Beats series) using Pytorch for inventory predictions, achieving 30% improvement over the naïve forecast. Applied **Agile** principles to iteratively refine data acquisition processes and update models, resulting in a 25% reduction in development cycle time.

Temple University: PhD Student in Quantitative Marketing Sep 2021 - May 2023

A study on the sentiment shift from live reactions (streaming comments) to reflection (reviews).

- Analyzed the evolution of sentiment from real-time streaming comments to post-consumption reviews, examining consistency in sentiment distribution, emerging topics, and emotional intensity.
- Applied **machine learning** and **transformer-based** models for topic modeling and sentiment analysis, uncovering how real-time excitement may contradict with long-term user satisfaction.

Projects

Movie Recommendation System | *Link* Deep Learning Java/Scala Web

"VideoRecSys", a movie recommendation system with a personalized user experience.

- Designed a three-page frontend with a main page ranking movies by historical ratings, a detailed page showing similar movies via embedding-based content, and a "Guess You Like" page based on user interactions.
- Developed three layers (recall, filtering, and re-ranking) for personalized content. Enabled dynamic adjustments using **XGboost** and **NeuralCF in dual-tower architecture**.
- Followed a robust deep learning architecture, integrating offline data processing on **Spark** and online model serving using **TensorFlow Serving** and **MLeap**.

AsyncTask Scheduler | Collaborated with *Hui Zhou* | *Link* Distributed Go Framework

"Async4Ai", a light-weighted framework asynchronously processing interdependent tasks.

- Designed a distributed task-scheduling system that decouples producers from consumers, highlighting complete task management via web module for interaction between agents and database.
- Crafted loosely coupled table normalization & sharding logic in **MySQL** to improve database performance.
- Performed stress testing and improved performance from 2k QPS to 10k QPS by using an efficient caching mechanism and distributed locks with **Redis** and reconfiguring the database connection pool.

Search Engine for Emojis | *Link* LLM-RAG Go/Python Web

"Emoji Search", an emoji search engine powered with AI enhanced filter.

- Adopted an event-driven architecture (**EDA**) for smoother search experience with four components.
- Streamed user search queries through **Kafka**. Optimized microservice discovery and balanced queries, reducing 30% service failures and 15% response times at 5000 QPS.
- Enhanced search results by leveraging a prompted Chatgpt built on a **RAG** pipeline to retrieve and refine candidates from an external emoji database. Containerized it in **Docker** and deployed it on **AWS EKS**.

TECHNICAL SKILLS

Languages: Python(5yrs), Java, Golang, c/C++, JavaScript, SQL, Scala; MySQL, Redis, DynamoDB, MongoDB
Frameworks: CUDA, Dask, PyTorch, Flask, Spring Boot, Express, Node, React, Angular, FastAPI, Django
Tools: Maven, JUnit, Postman, Terraform, Nginx, CI/CD, RabbitMQ, Kubernetes, Kafka, Docker, Spark
Miscellaneous: Linux, Git, AWS(*Certified Solutions Architect Associate*), Azure (*Certified AI Engineer Associate*)